Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

Zheng, Chiang, and Sheng et al. (2023)

Hins Fang Charles Costanzo

The models: LLaMA and Vicuna



LLaMA-13B (Meta, 2023)



Vicuna-13B (LMSYS, 2023)

Photos: PA (llama) and Marshallhenrie (vicuña), Wikimedia Commons, CC BY-SA 4.0

Why “LLaMA” and “Vicuna”?

Where do the names come from?

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LLaMA is a backronym: **L**arge **La**nguage **M**odel Meta **A**I

Vicuna is a fine-tuned LLaMA from LMSYS, named after a relative of the llama:

- Llama → Meta’s base model
- Alpaca → Stanford’s fine-tune
- Vicuna → LMSYS’s fine-tune, trained on ShareGPT conversations

The vicuña is believed to be the ancestor of the domesticated alpaca.

The end of LLaMA? (2023–2026)

Feb 2023	LLaMA released to researchers; weights leak via BitTorrent [1]
Mar 2023	Vicuna: LMSYS fine-tunes LLaMA on ShareGPT conversations
Jun 2023	**MT-Bench and Chatbot Arena ⟵ Zheng et al. [2]**
Jul 2023	LLaMA 2 [3]
Apr 2024	LLaMA 3 [4]
Apr 2025	LLaMA 4 [5]
Apr 2026	**Muse Spark** [6]
Aug 2026	**Muse Glimmer:** open weights, pitched for “LLM-as-a-judge” [7]
Oct 2026	**Muse Spark Open** [8]

Example 1:

Question

If the FED buys bonds in the secondary market

(A) the money supply will increase.
(B) the money supply will decrease.
(C) the money supply will not be affected.
(D) the discount rate would be affected.

Output the answer only.

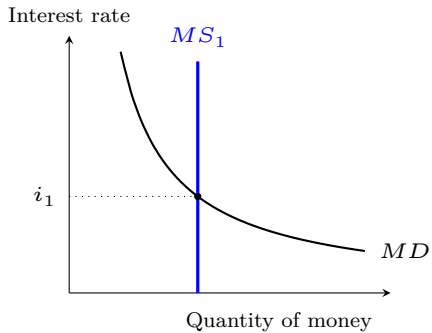
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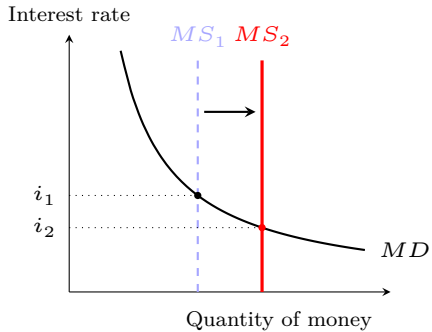
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How did they do?

LLaMA-13B

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The Federal Reserve buys bonds in the secondary
market to increase the money supply.
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The prompt said to "output the answer only."

- LLaMA provided its own answer.
- Vicuna outputted the correct answer verbatim from the four options listed.

Example 1 - Follow-up:

Follow-up Question

```
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LLaMA-13B

The Federal Reserve buys bonds in the secondary market to increase the money supply. This affects your daily life by:

1. Increasing the money supply.
2. Increasing the money supply and lowering the discount rate.
3. Increasing the money supply and lowering the discount rate and increasing the money supply.

Vicuna-13B

When the Federal Reserve buys bonds in the secondary market, it can have several effects on the economy that can impact your daily life. Here are three examples:

1. Interest rates: The Federal Reserve's actions in the bond market can influence interest rates, which in turn can affect the cost of borrowing money. For example, if the Federal Reserve buys bonds and drives up demand for them, interest rates on things like mortgages, car loans, and credit cards may go up, making it more expensive to borrow money.
2. Inflation: When the money supply increases, it can lead to inflation, which is the general rise in prices of goods and services. This can impact your daily life by making it more expensive to buy things like groceries, gas, and other necessities.
3. Employment: The Federal Reserve's actions in the bond market can also have an impact on employment. For example, if the Federal Reserve's actions lead to economic growth and increased demand for goods and services, it can create more job opportunities and boost employment.

Example 1 - GPT-4 as judge

GPT-4 Judgment

LLaMA-13B provided an incorrect response to the user's question about how the Federal Reserve buying bonds in the secondary market affects daily life. The answer given is repetitive and lacks clear examples of how the action impacts daily life.

On the other hand, Vicuna-13B provided a relevant and accurate response to the user's question about the Federal Reserve buying bonds. The answer includes three clear examples of how the action impacts daily life, such as interest rates, inflation, and employment.

LLaMA-13B should improve by providing relevant, clear, and accurate examples to address the user's question. **Vicuna-13B's response is more helpful, relevant, and detailed.**

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Vicuna more helpful, relevant, and detailed

- LLaMA: repetitive, no real examples
- Vicuna: three clear, accurate examples

Scores under different benchmarks

Example 1 is an MMLU question.

Model	MMLU (5-shot)	MT-Bench score (GPT-4)
LLaMA-13B	47.0	2.61
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Table 8 in the paper. MT-Bench is scored 1–10 by GPT-4.

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Table 8 in the paper. MT-Bench is scored 1–10 by GPT-4.

- On MMLU: a 5-point gap
- On human-style preference: Vicuna scores **2.4×** higher

These benchmarks were designed to measure knowledge, not human preferences.

Key terms

Base model Pre-trained on raw text to predict the next token (LLaMA)

Instruction fine-tuning Further training on (instruction, response) pairs so it follows directions (Vicuna)

RLHF Reinforcement learning from human feedback: humans rank outputs, the model is tuned toward the winners (GPT-4, Claude)

Aligned model A base model after either of the above

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Users strongly prefer aligned models, but *measuring* that is not straightforward.

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- Open-ended answers lack a reference solution to check against
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Research question: can a "strong'' LLM stand in for the human raters?

LLM-as-a-judge

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Check for:

1. What **biases** does the judge have, and how can we mitigate them? (Section 3)
2. Does the judge **agree with humans**? (Section 4)

Paper's contribution

1. Studies LLM judges directly, including prompting, LLM biases, and how to mitigate some of them
2. Two human-preference datasets to validate against:
   - **MT-Bench**: 80 multi-turn questions, 3K expert votes
   - **Chatbot Arena**: crowdsourced anonymous battles, 30K conversations
3. Main result: GPT-4 agrees with humans over **80%** of the time, the same rate humans agree with each other

Outline

1. Introduction — *Charles*
2. MT-Bench and Chatbot Arena — *Charles*
3. LLM as a judge: types, biases, fixes — *Charles*
4. Agreement with humans — *Hins*
5. Human preference vs. standard benchmarks — *Hins*
6. Discussion and limitations — *Hins*

References

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